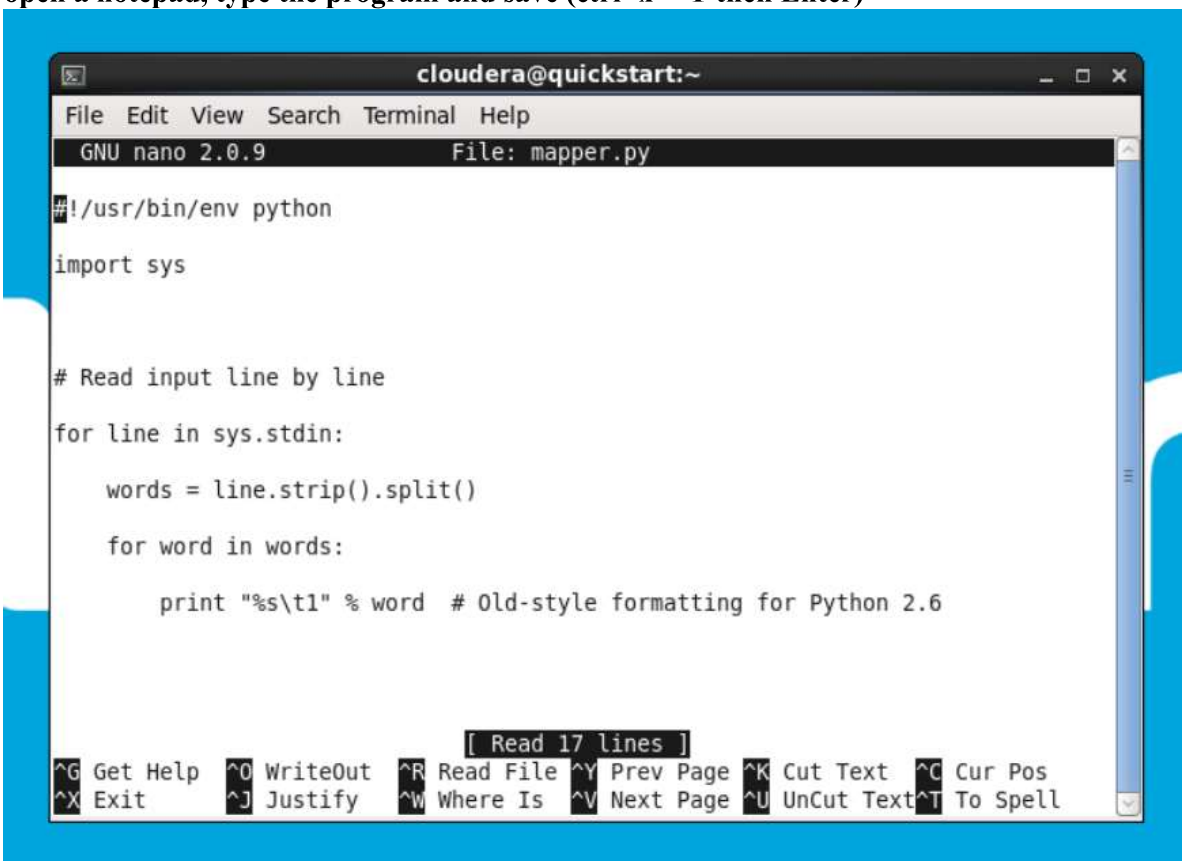
TASK 3

a) Write a word count program to simulate the Map Reduce Paradigm.
b) Write a Map Reduce program to perform matrix multiplication.

A) Write a word count program to simulate the Map Reduce Paradigm.

Create mapper.py
nano mapper.py

open a notepad, type the program and save (ctrl+x-> Y then Enter)



Mapper.py code

```
#!/usr/bin/env python
import sys

# Read input line by line
for line in sys.stdin:
    words = line.strip().split()
    for word in words:
        print "%s\t1" % word
```

Execute mapper.py

```
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ at /user/cloudera/wordcount/output/part-00000

syntax error. Last token seen: /
Garbled time
[cloudera@quickstart ~]$
[cloudera@quickstart ~]$ hdfs dfs -ls /user/cloudera/wordcount/output

Found 2 items
-rw-r--r--   1 cloudera cloudera          0 2025-02-12 07:16 /user/cloudera/word
count/output/_SUCCESS
-rw-r--r--   1 cloudera cloudera         47 2025-02-12 07:16 /user/cloudera/word
count/output/part-00000
[cloudera@quickstart ~]$
[cloudera@quickstart ~]$ hdfs dfs -cat /user/cloudera/wordcount/output/part-0000
0
Bharath 3
Hai     1
Hello   1
Yannam  1
hai     1
hello   1
[cloudera@quickstart ~]$ nano mapper.py
[cloudera@quickstart ~]$ chmod +x mapper.py
[cloudera@quickstart ~]$
```

Test using

echo "hello world hello" | python mapper.py

```
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help

Found 2 items
-rw-r--r--   1 cloudera cloudera          0 2025-02-12 07:16 /user/cloudera/word
count/output/_SUCCESS
-rw-r--r--   1 cloudera cloudera         47 2025-02-12 07:16 /user/cloudera/word
count/output/part-00000
[cloudera@quickstart ~]$
[cloudera@quickstart ~]$ hdfs dfs -cat /user/cloudera/wordcount/output/part-0000
0
Bharath 3
Hai     1
Hello   1
Yannam  1
hai     1
hello   1
[cloudera@quickstart ~]$ nano mapper.py
[cloudera@quickstart ~]$ chmod +x mapper.py
[cloudera@quickstart ~]$ echo "hello world hello" | python mapper.py

hello   1
world   1
hello   1
[cloudera@quickstart ~]$
[cloudera@quickstart ~]$
```

Create reducer.py
nano mapper.py

Reducer.py Code:

```
#!/usr/bin/env python
import sys
word_counts = {}
# Read input from stdin
for line in sys.stdin:
    word, count = line.strip().split("\t", 1)  # Split by tab
    try:
        count = int(count)  # Convert count to integer
        if word in word_counts:
            word_counts[word] += count
        else:
            word_counts[word] = count
    except ValueError:
        pass  # Ignore lines with unexpected format

# Print word counts
for word in sorted(word_counts):
    print "%s\t%d" % (word, word_counts[word])
```

```
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
  GNU nano 2.0.9                File: reducer.py

#!/usr/bin/env python

import sys


word_counts = {}


# Reom stdin

for line in sys.stdin:

    word, count = line.strip().split("\t", 1)  # Split by tab

    try:

        count = int(count)  # Convert count to integer
                                [ Read 41 lines ]
^G Get Help  ^O WriteOut  ^R Read File ^Y Prev Page ^K Cut Text  ^C Cur Pos
^X Exit      ^J Justify   ^W Where Is  ^V Next Page ^U UnCut Text^T To Spell
```

execute using

chmod +x reducer.py

```
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
Found 2 items
-rw-r--r--   1 cloudera cloudera          0 2025-02-12 07:16 /user/cloudera/word
count/output/_SUCCESS
-rw-r--r--   1 cloudera cloudera         47 2025-02-12 07:16 /user/cloudera/word
count/output/part-00000
[cloudera@quickstart ~]$
[cloudera@quickstart ~]$ hdfs dfs -cat /user/cloudera/wordcount/output/part-0000
0
Bharath 3
Hai     1
Hello   1
Yannam  1
hai     1
hello   1
[cloudera@quickstart ~]$ nano mapper.py
[cloudera@quickstart ~]$ chmod +x mapper.py
[cloudera@quickstart ~]$ echo "hello world hello" | python mapper.py

hello   1
world   1
hello   1
[cloudera@quickstart ~]$
[cloudera@quickstart ~]$ nano reducer.py
[cloudera@quickstart ~]$ chmod +x reducer.py
```

Test Using:

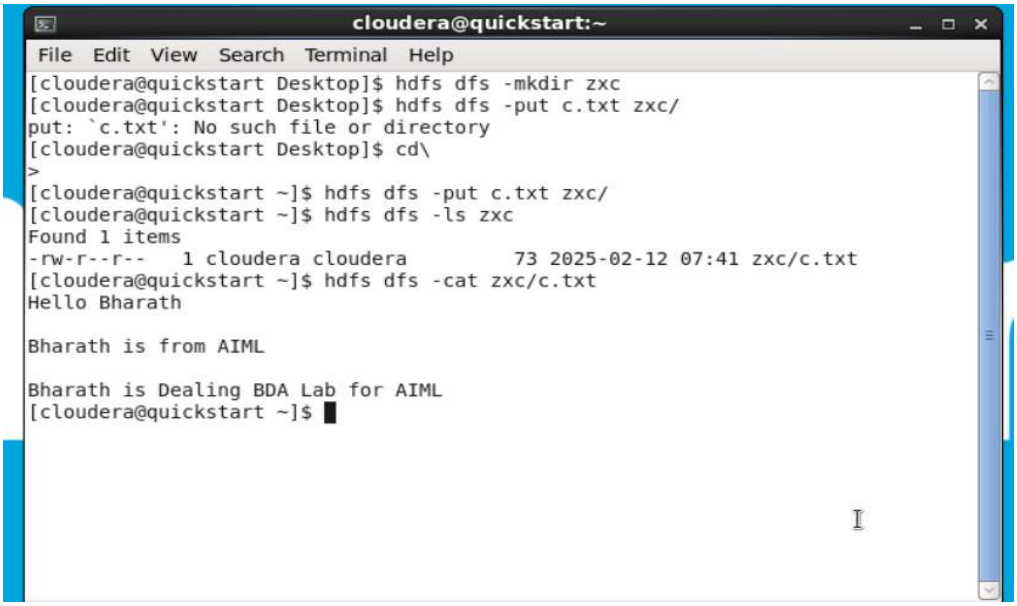
echo -e "hello\t1\nworld\t1\nhello\t1" | python reducer.py

```
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ hdfs dfs -cat /user/cloudera/wordcount/output/part-0000
0
Bharath 3
Hai     1
Hello   1
Yannam  1
hai     1
hello   1
[cloudera@quickstart ~]$ nano mapper.py
[cloudera@quickstart ~]$ chmod +x mapper.py
[cloudera@quickstart ~]$ echo "hello world hello" | python mapper.py

hello   1
world   1
hello   1
[cloudera@quickstart ~]$
[cloudera@quickstart ~]$ nano reducer.py
[cloudera@quickstart ~]$ chmod +x reducer.py
[cloudera@quickstart ~]$ echo -e "hello\t1\nworld\t1\nhello\t1" | python reducer
.py
hello   2
world   1
[cloudera@quickstart ~]$
[cloudera@quickstart ~]$
```

cretae a text file

```
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
Hello Bharath

Bharath is from AIML

Bharath is Dealing BDA Lab for AIML
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
"c.txt" 5L, 73C                                    5,1           All
```

Export to HDFS:



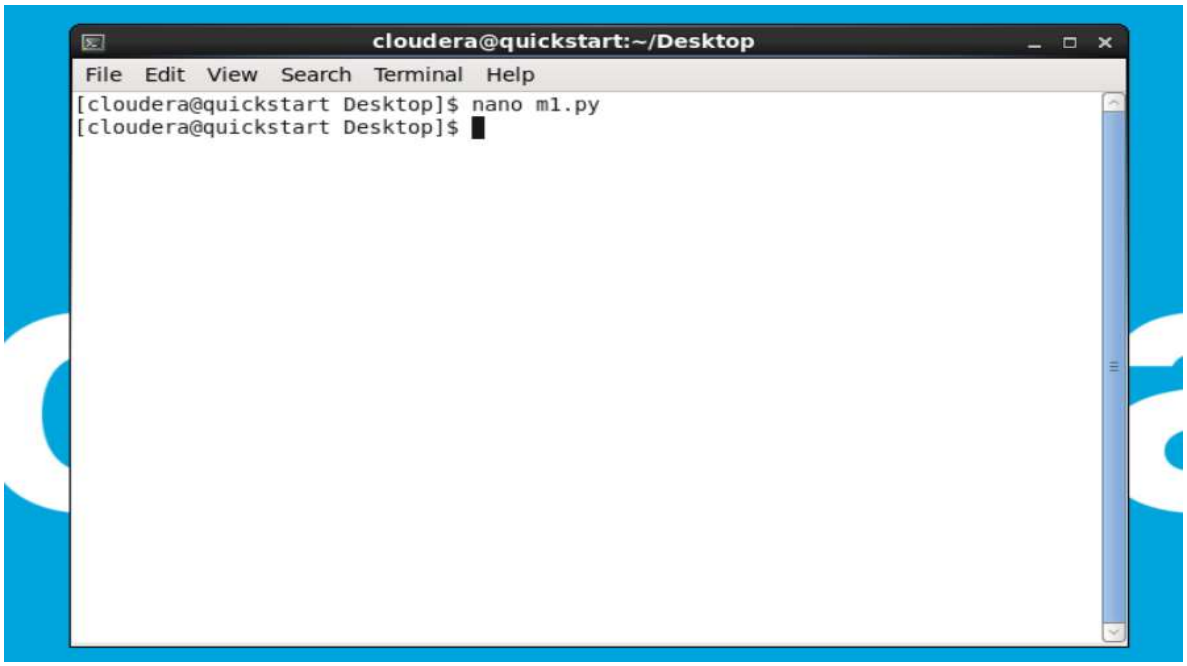
Run the Command

hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar -input zxc/c.txt -output zxc/output -mapper "python mapper.py" -reducer "python reducer.py" -file mapper.py -file reducer.py

```
[cloudera@quickstart ~]$ hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.j
ar -input zxc/c.txt -output zxc/output -mapper "python mapper.py" -reducer "pyth
on reducer.py" -file mapper.py -file reducer.py
25/02/12 07:47:42 WARN streaming.StreamJob: -file option is deprecated, please u
se generic option -files instead.
packageJobJar: [mapper.py, reducer.py] [/usr/lib/hadoop-mapreduce/hadoop-streami
ng-2.6.0-cdh5.13.0.jar] /tmp/streamjob6447859711204783167.jar tmpDir=null
25/02/12 07:47:45 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0
:8032
25/02/12 07:47:46 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0
:8032
25/02/12 07:47:48 INFO mapred.FileInputFormat: Total input paths to process : 1
25/02/12 07:47:49 INFO mapreduce.JobSubmitter: number of splits:2
25/02/12 07:47:50 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_17
39366319868_0002
25/02/12 07:47:51 INFO impl.YarnClientImpl: Submitted application application_17
39366319868_0002
25/02/12 07:47:51 INFO mapreduce.Job: The url to track the job: http://quickstar
t.cloudera:8088/proxy/application_1739366319868_0002/
25/02/12 07:47:51 INFO mapreduce.Job: Running job: job_1739366319868_0002
25/02/12 07:47:58 INFO mapreduce.Job: Job job_1739366319868_0002 running in uber
 mode : false
25/02/12 07:47:58 INFO mapreduce.Job:  map 0% reduce 0%
25/02/12 07:48:09 INFO mapreduce.Job:  map 100% reduce 0%
25/02/12 07:48:19 INFO mapreduce.Job:  map 100% reduce 100%
25/02/12 07:48:20 INFO mapreduce.Job: Job job_1739366319868_0002 completed succe
ssfully
25/02/12 07:48:21 INFO mapreduce.Job: Counters: 49
        File System Counters
                FILE: Number of bytes read=129
                FILE: Number of bytes written=441866
```

Output file is created in zxc directory

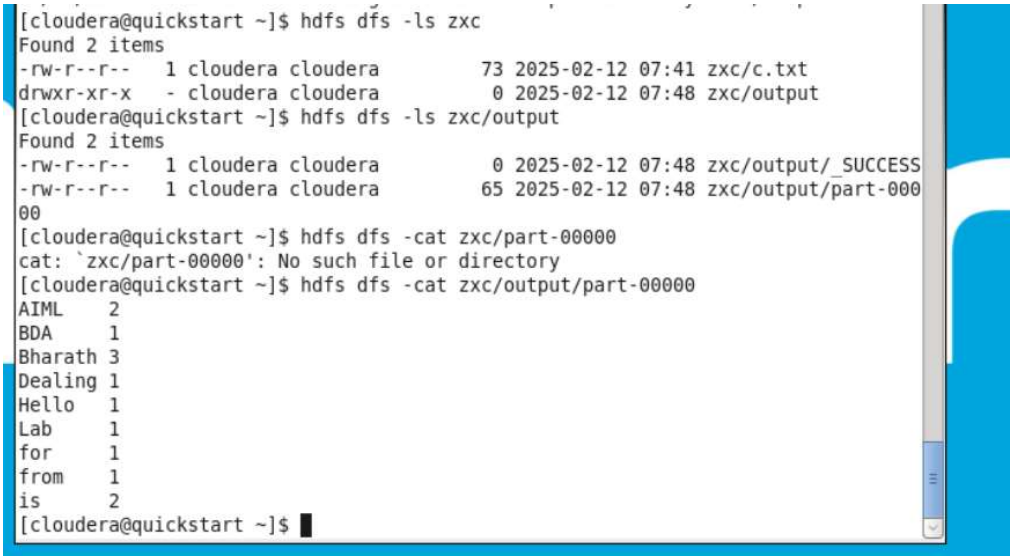
```
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=225
                CPU time spent (ms)=5880
                Physical memory (bytes) snapshot=766599168
                Virtual memory (bytes) snapshot=4762079232
                Total committed heap usage (bytes)=790102016
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=110
        File Output Format Counters
                Bytes Written=65
25/02/12 07:48:21 INFO streaming.StreamJob: Output directory: zxc/output
[cloudera@quickstart ~]$ hdfs dfs -ls zxc
Found 2 items
-rw-r--r--   1 cloudera cloudera         73 2025-02-12 07:41 zxc/c.txt
drwxr-xr-x   - cloudera cloudera          0 2025-02-12 07:48 zxc/output
[cloudera@quickstart ~]$
```



Run the output

hdfs dfs -ls zxc/outut

hdfs dfs -cat zxc/output/part-00000



B) Write a Map Reduce program to perform matrix multiplication.

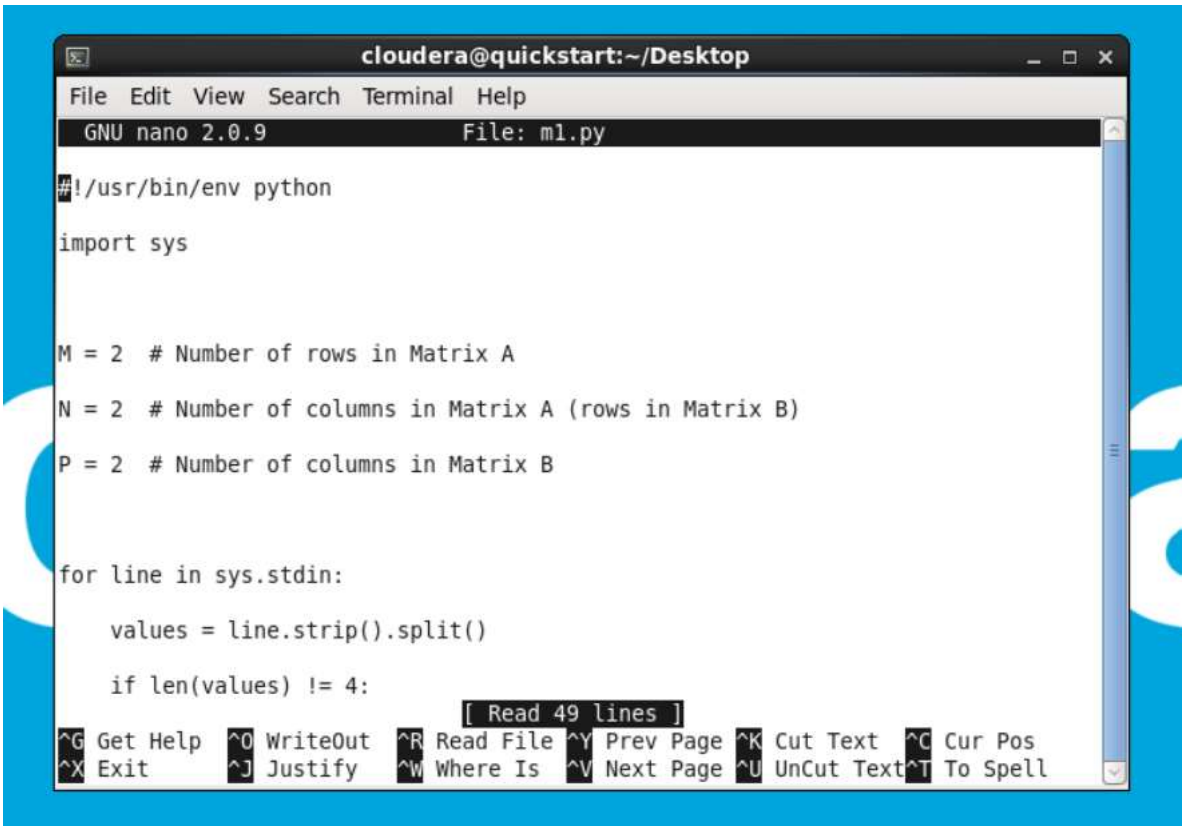
1. Create a file for mapper

nano m1.py

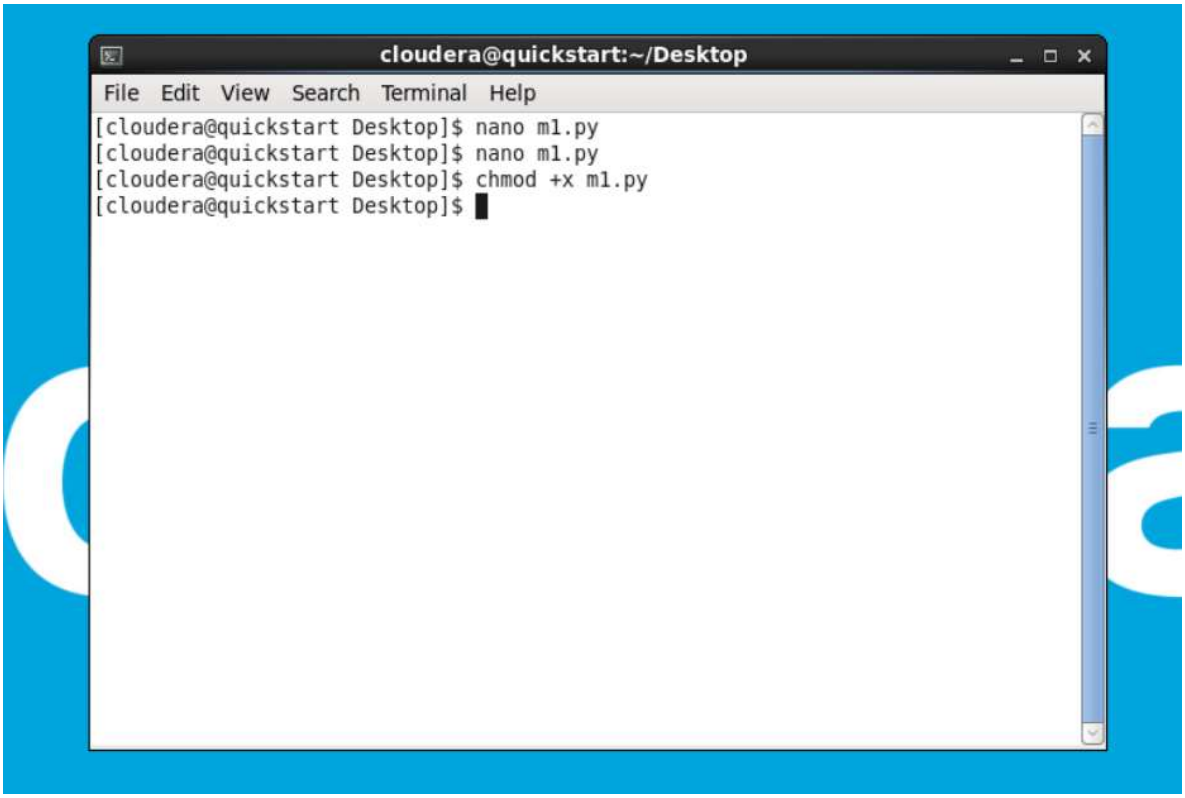
2. write mapper class code:

mapper.py:

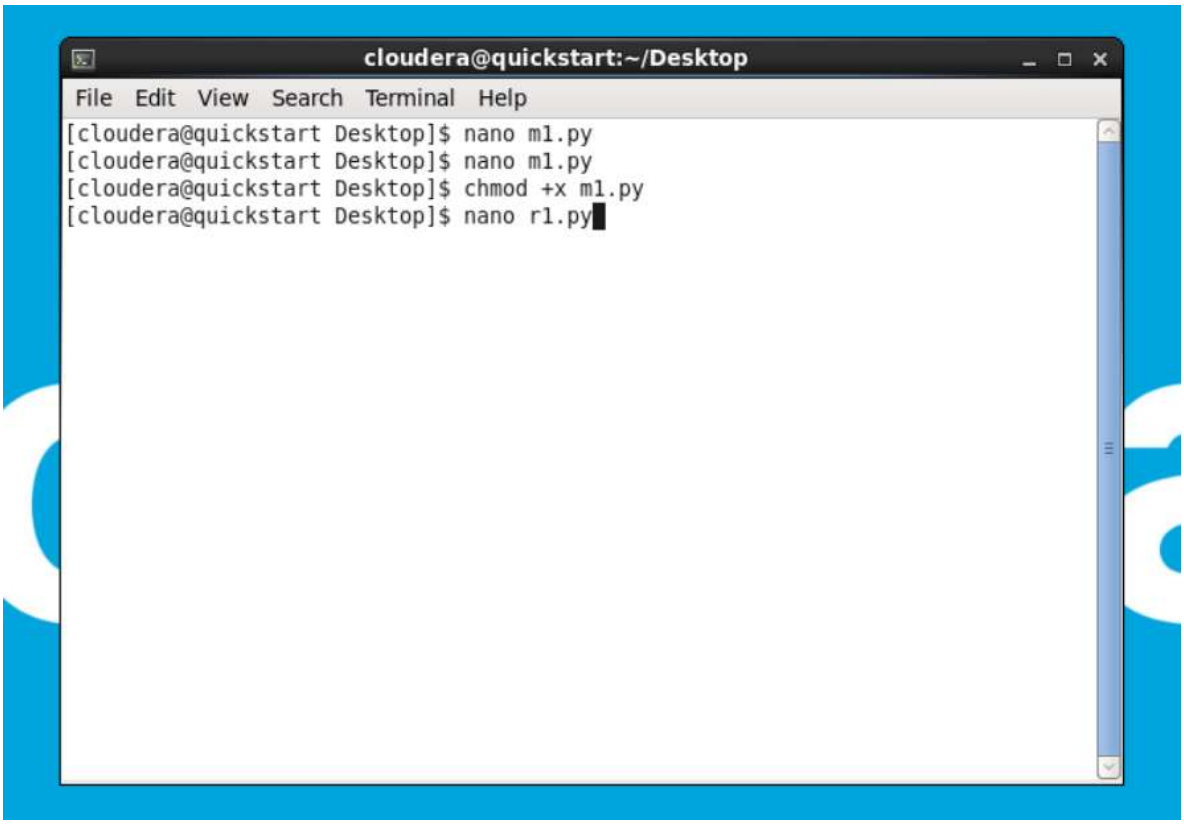
```
#!/usr/bin/env python
import sys

M = 2  # Number of rows in Matrix A
N = 2  # Number of columns in Matrix A (also rows in Matrix B)
P = 2  # Number of columns in Matrix B

for line in sys.stdin:
    values = line.strip().split()
    if len(values) != 4:
        continue  # Skip malformed lines

    matrix, row, col, val = values
    row = int(row)
    col = int(col)
    val = float(val)

    if matrix == "A":
        for k in range(P):  # P is the number of columns in Matrix B
            print "%d,%d\tA,%d,%f" % (row, k, col, val)

    elif matrix == "B":
        for i in range(M):  # M is the number of rows in Matrix A
            print "%d,%d\tB,%d,%f" % (i, col, row, val)
```



3. make the code executable:
   chmod +x m1.py



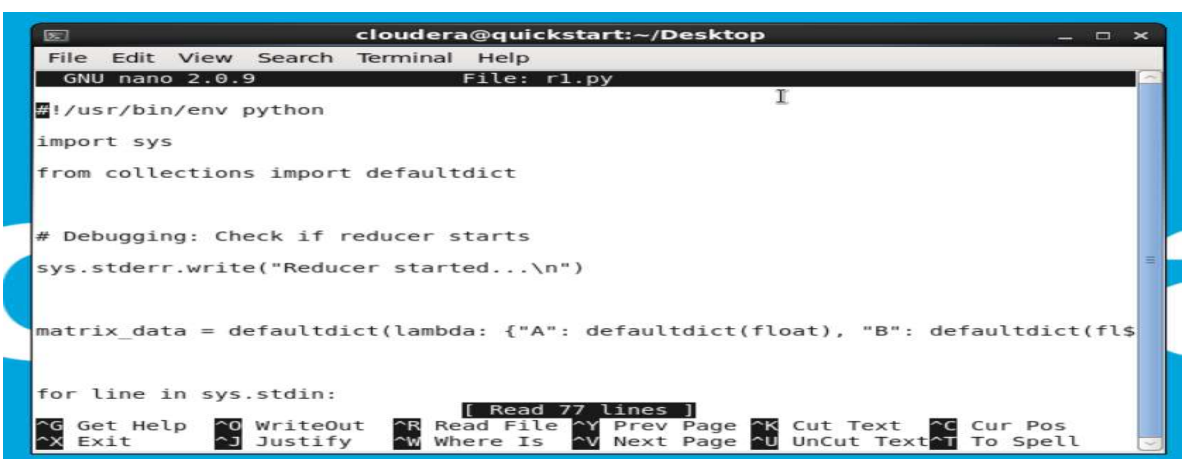
4. write the code for reducer class:
   nano r1.py



reducer.py

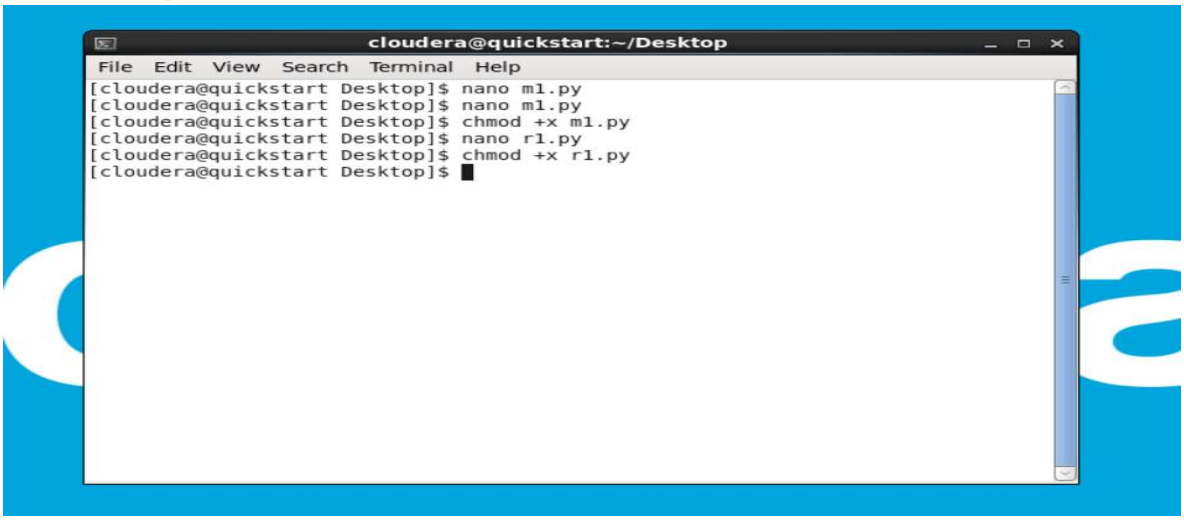
```
#!/usr/bin/env python
import sys

# Debugging: Check if reducer starts
sys.stderr.write("Reducer started...\n")

# Dictionary to store matrix values
matrix_data = {}

# Read input from mapper
for line in sys.stdin:
    sys.stderr.write("Received line: %s\n" % line)  # Debugging
    line = line.strip()

    if "\t" not in line:
        continue  # Skip malformed lines

    try:
        key, value = line.split("\t")
        i, j = map(int, key.split(","))
```

```
        parts = value.split(",")
        if len(parts) != 3:
            continue  # Skip incorrectly formatted lines

        matrix_type = parts[0]
        k = int(parts[1])
        val = float(parts[2])

        if (i, j) not in matrix_data:
            matrix_data[(i, j)] = {"A": {}, "B": {}}

        if matrix_type == "A":
            matrix_data[(i, j)]["A"][k] = val
        elif matrix_type == "B":
            matrix_data[(i, j)]["B"][k] = val

    except Exception as e:
        sys.stderr.write("Error processing line: %s\n" % line)
        sys.stderr.write("Exception: %s\n" % str(e))
        continue  # Skip problematic lines

# Compute matrix multiplication result
for (i, j), values in matrix_data.items():
    result = 0
    a_values = values["A"]
    b_values = values["B"]

    for k in a_values:
        if k in b_values:
            result += a_values[k] * b_values[k]

    print "%d,%d\t%f" % (i, j, result)
```



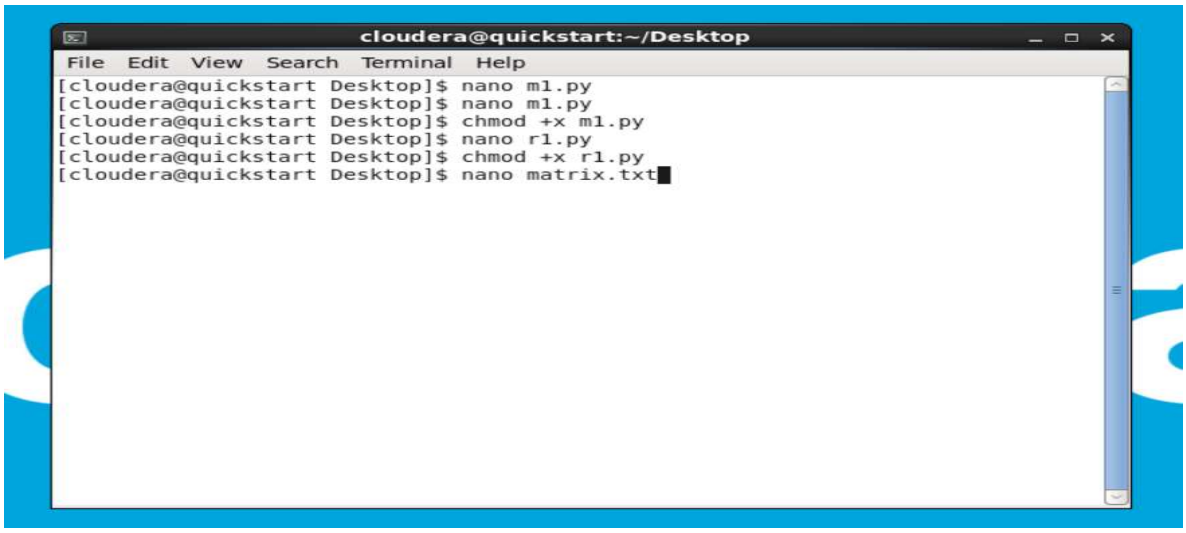
5. make it executable:

chmod +x r1.py



6. create a text file for matrix input

   nano matrix.txt



Write the matrix input (2X2 matrix)

A 0 0 1
A 0 1 2
A 1 0 3
A 1 1 4
B 0 0 5
B 0 1 6
B 1 0 7

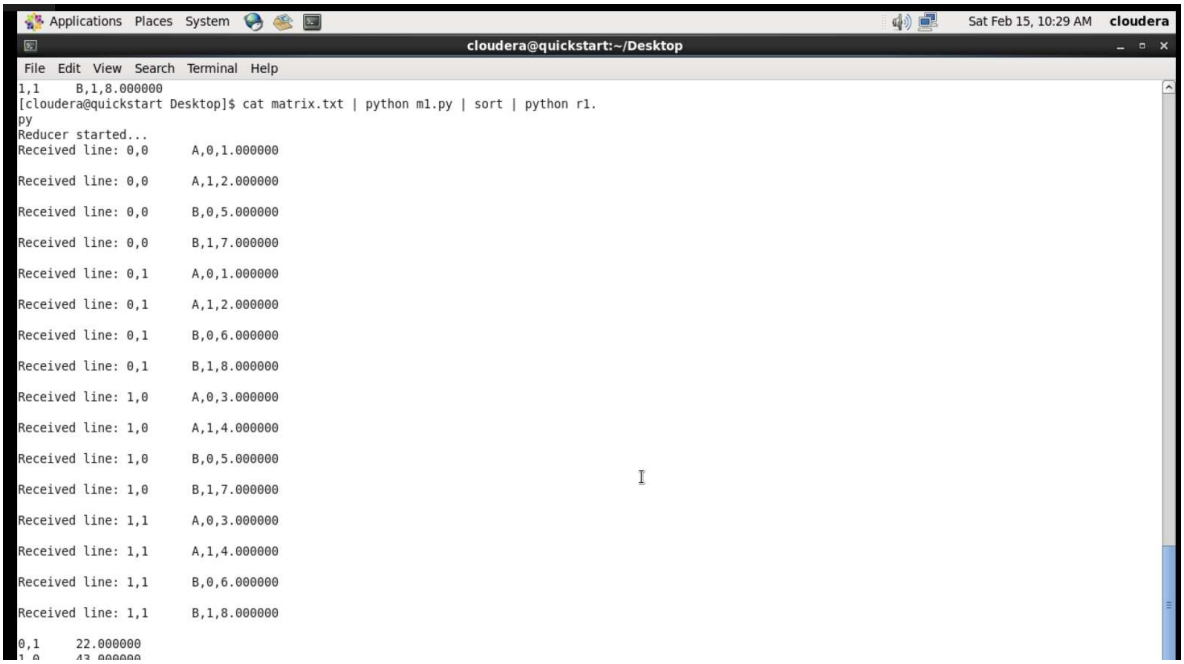
B 1 1 8

```
cloudera@quickstart:~/Desktop
File  Edit  View  Search  Terminal  Help
  GNU nano 2.0.9              File: matrix.txt

A   0   0   1

A   0   1   2

A   1   0   3

A   1   1   4

B   0   0   5

B   0   1   6

B   1   0   7

B   1   1   8

                          [ Read 15 lines ]
^G Get Help  ^O WriteOut  ^R Read File ^Y Prev Page ^K Cut Text  ^C Cur Pos
^X Exit      ^J Justify   ^W Where Is  ^V Next Page ^U UnCut Text^T To Spell
```

Run the mapper class: cat matrix.txt | python m1.py

```
cloudera@quickstart:~/Desktop
File  Edit  View  Search  Terminal  Help
0,0     19.000000
1,1     50.000000
[cloudera@quickstart Desktop]$ cat matrix/matrix.txt | python m.py
cat: matrix/matrix.txt: No such file or directory
python: can't open file 'm.py': [Errno 2] No such file or directory
[cloudera@quickstart Desktop]$
[cloudera@quickstart Desktop]$ cat matrix.txt | python m1.py
0,0     A,0,1.000000
0,1     A,0,1.000000
0,0     A,1,2.000000
0,1     A,1,2.000000
1,0     A,0,3.000000
1,1     A,0,3.000000
1,0     A,1,4.000000
1,1     A,1,4.000000
0,0     B,0,5.000000
1,0     B,0,5.000000
0,1     B,0,6.000000
1,1     B,0,6.000000
0,0     B,1,7.000000
1,0     B,1,7.000000
0,1     B,1,8.000000
1,1     B,1,8.000000
[cloudera@quickstart Desktop]$
```

Run the reducer class: cat matrix.txt | python m1.py | sort | python r1.py

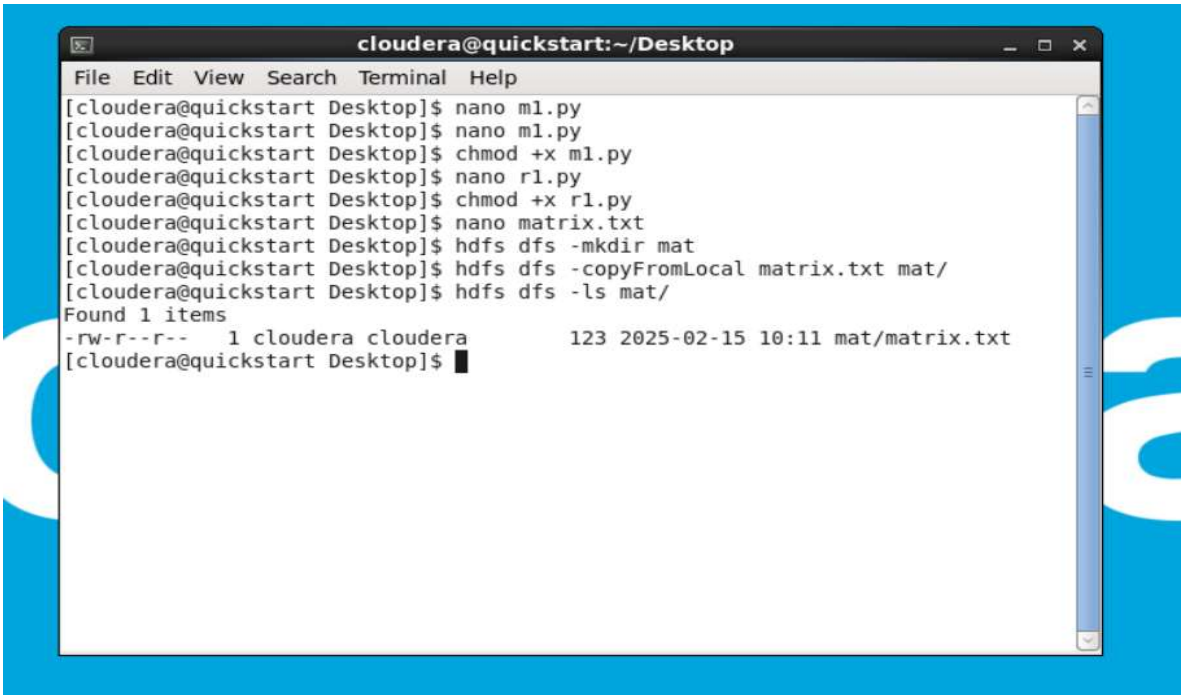


7. upload this to hdfs

   hdfs dfs -mkdir mat

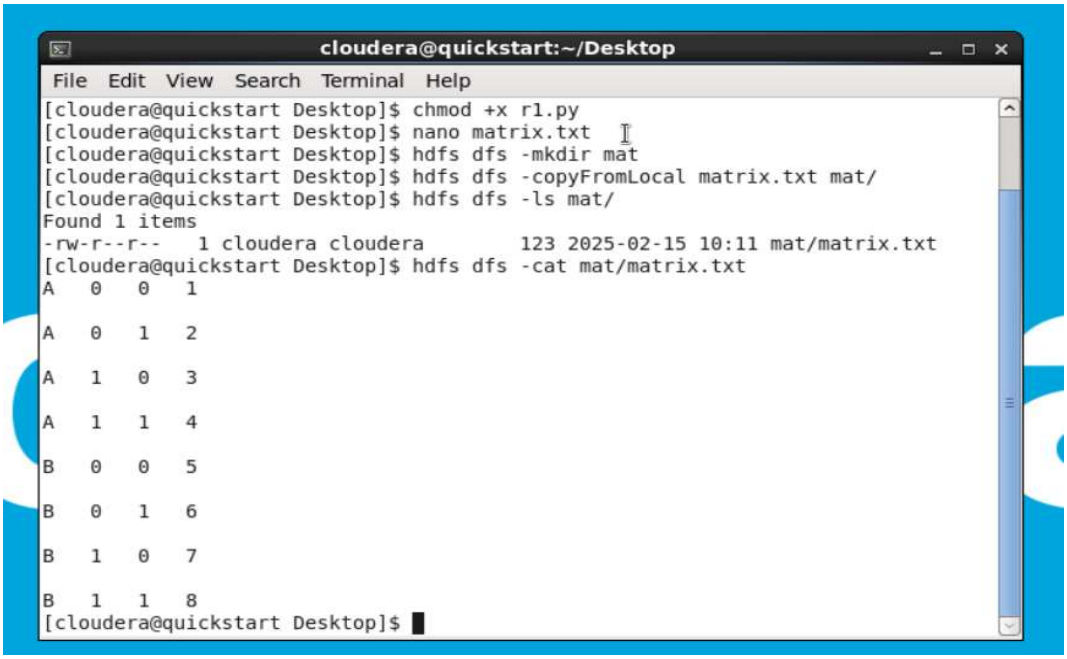
   hdfs dfs -copyFromLocal matrix.txt mat/

   hdfs dfs -ls mat/



Read the input file in hdfs

Hdfs dfs -cat mat/matrix.txt



8. Run the input file to perform matrix multiplication using mapper and reducer classes

hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar -input mat/matrix.txt -output mat/output -mapper "python m1.py" -reducer "python r1.py" -file m1.py -file r1.py

```
cloudera@quickstart:~/Desktop
File  Edit  View  Search  Terminal  Help
                Reduce input records=16
                Reduce output records=4
                Spilled Records=32
                Shuffled Maps =2
                Failed Shuffles=0
                Merged Map outputs=2
                GC time elapsed (ms)=281
                CPU time spent (ms)=3900
                Physical memory (bytes) snapshot=730992640
                Virtual memory (bytes) snapshot=4742799360
                Total committed heap usage (bytes)=773849088
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=185
        File Output Format Counters
                Bytes Written=56
25/02/15 10:18:52 INFO streaming.StreamJob: Output directory: mat/output
[cloudera@quickstart Desktop]$
```

9. Open the output file in mat directory
   hdfs dfs -ls mat
   hdfs dfs -ls mat/output

```
cloudera@quickstart:~/Desktop
File  Edit  View  Search  Terminal  Help
                Virtual memory (bytes) snapshot=4742799360
                Total committed heap usage (bytes)=773849088
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=185
        File Output Format Counters
                Bytes Written=56
25/02/15 10:18:52 INFO streaming.StreamJob: Output directory: mat/output
[cloudera@quickstart Desktop]$ hdfs dfs -ls mat
Found 2 items
-rw-r--r--   1 cloudera cloudera        123 2025-02-15 10:11 mat/matrix.txt
drwxr-xr-x   - cloudera cloudera          0 2025-02-15 10:18 mat/output
[cloudera@quickstart Desktop]$ hdfs dfs -ls mat/output
Found 2 items
-rw-r--r--   1 cloudera cloudera          0 2025-02-15 10:18 mat/output/_SUCCESS
-rw-r--r--   1 cloudera cloudera         56 2025-02-15 10:18 mat/output/part-000
00
[cloudera@quickstart Desktop]$
```

10. Open the part file to see the output
    hdfs dfs -cat mat/output/part-00000

```
cloudera@quickstart:~/Desktop
File  Edit  View  Search  Terminal  Help
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=185
        File Output Format Counters
                Bytes Written=56
25/02/15 10:18:52 INFO streaming.StreamJob: Output directory: mat/output
[cloudera@quickstart Desktop]$ hdfs dfs -ls mat
Found 2 items
-rw-r--r--   1 cloudera cloudera        123 2025-02-15 10:11 mat/matrix.txt
drwxr-xr-x   - cloudera cloudera          0 2025-02-15 10:18 mat/output
[cloudera@quickstart Desktop]$ hdfs dfs -ls mat/output
Found 2 items
-rw-r--r--   1 cloudera cloudera          0 2025-02-15 10:18 mat/output/_SUCCESS
-rw-r--r--   1 cloudera cloudera         56 2025-02-15 10:18 mat/output/part-000
00
[cloudera@quickstart Desktop]$ hdfs dfs -cat mat/output/part-00000
0,1     22.000000
1,0     43.000000
0,0     19.000000
1,1     50.000000
[cloudera@quickstart Desktop]$
```